

Agentic Accessibility Auditor

Accessibility Guidelines & Rule Mapping

Internship Project — FAST University 2025

1. All 30 Accessibility Guidelines

Guidelines G01–G10 are from the original project internship plan. Guidelines G11–G30 cover deaf users, color-blind users, motor-impaired users, cognitive disabilities, and elderly users.

ID	Issue / Guideline	Description	User Group	WCAG
G01	Missing accessible label	Every interactive/clickable element must have a visible or programmatic label so screen readers can announce it.	Blind / Low Vision	WCAG 4.1.2
G02	Image button without description	ImageButton and clickable ImageView elements must carry a content description that conveys their purpose.	Blind	WCAG 1.1.1
G03	Duplicate labels	No two interactive elements on the same screen should share identical visible text or content descriptions unless they perform the same action.	Blind / Cognitive	WCAG 4.1.2
G04	Small touch target	All tappable elements must meet minimum size requirements so that users with motor impairments can activate them reliably.	Motor Impaired / Elderly	WCAG 2.5.5
G05	Unlabeled input field	Every text input field must have a programmatic label (hint, labelFor, or content-desc) so screen readers can identify it.	Blind / Cognitive	WCAG 1.3.1
G06	Disabled important control	Controls that are critical to the user's task must not be permanently disabled or hidden without a clear explanation or alternative.	All Users	WCAG 2.1.1
G07	Invisible or zero-size component	Elements with zero visible area or invalid bounds must be removed from the layout and accessibility focus tree.	All Users / Blind	WCAG 1.3.1
G08	Possible layout overlap	UI elements must not overlap each other in ways that hide content or disrupt the focus traversal order.	All Users / Blind	WCAG 1.3.2
G09	Low contrast	Text and meaningful UI components must maintain sufficient contrast ratio against their background for readability.	Low Vision / Color Blind	WCAG 1.4.3

G10	Text overflow	Text component bounds must be large enough to display their full content without clipping or truncation.	<i>Low Vision / Cognitive</i>	WCAG 1.4.4
G11	Information conveyed by color alone	Color must never be the only means of conveying information, indicating an action, or distinguishing a visual element.	<i>Color Blind</i>	WCAG 1.4.1
G12	Missing captions on video content	All video content that includes spoken audio must provide synchronized captions or subtitles.	<i>Deaf / Hard of Hearing</i>	WCAG 1.2.2
G13	Audio-only content without transcript	Any content delivered only through audio must have an equivalent text transcript available in the UI.	<i>Deaf / Hard of Hearing</i>	WCAG 1.2.1
G14	Notification uses audio only	Alerts and notifications must use visual indicators in addition to any audio alert so deaf users are informed.	<i>Deaf / Hard of Hearing</i>	WCAG 1.3.3
G15	Logical focus / reading order	The keyboard and screen reader focus order must follow a natural top-to-bottom, left-to-right reading sequence that matches the visual layout.	<i>Blind / Motor Impaired</i>	WCAG 1.3.2 / 2.4.3
G16	Non-interactive elements in focus tree	Decorative images, dividers, and purely visual elements must be excluded from the accessibility focus order.	<i>Blind</i>	WCAG 1.3.1
G17	Insufficient touch target spacing	Adjacent interactive elements must have enough spacing between them to prevent accidental activation by users with motor impairments.	<i>Motor Impaired</i>	WCAG 2.5.5
G18	Functionality requires multi-finger gesture	All app functionality must be achievable with a single pointer, with no feature locked behind multi-touch gestures.	<i>Motor Impaired / Single Hand</i>	WCAG 2.1.1
G19	No confirmation for destructive action	Actions that are irreversible or destructive must require a confirmation step to prevent accidental activation.	<i>Motor Impaired / Cognitive</i>	WCAG 3.3.4
G20	Input label disappears on focus	Input field labels must remain persistently visible even when the field is focused and being typed into.	<i>Cognitive / Memory Impaired</i>	WCAG 3.3.2
G21	Vague or missing error messages	Error messages must clearly identify which field has an error and provide specific guidance on how to correct it.	<i>Cognitive / Blind</i>	WCAG 3.3.1 / 3.3.3
G22	Password field lacks show/hide toggle	Password input fields must include an accessible toggle to reveal or hide the entered password to reduce typing errors.	<i>Cognitive / Motor Impaired</i>	WCAG 3.3.1
G23	Navigation control not labeled	Navigation elements such as back, home, close, and menu buttons must always be accessible and carry descriptive labels.	<i>Blind</i>	WCAG 2.4.6

G24	Screen has no descriptive title	Every screen must have a visible and descriptive title or heading so users know where they are within the application.	<i>Blind / Cognitive</i>	WCAG 2.4.2
G25	Uncontrolled auto-updating content	Animated or automatically updating content must provide controls to pause, stop, or hide the motion.	<i>Cognitive / Epilepsy / ADHD</i>	WCAG 2.2.2
G26	Session timeout without warning	If a session has a time limit, the user must be warned at least 20 seconds before expiry and given the option to extend.	<i>Cognitive / Slow Readers</i>	WCAG 2.2.1
G27	Complex or jargon-heavy labels	All visible text, labels, hints, and content descriptions must use plain, simple language that users with low literacy can understand.	<i>Cognitive / Low Literacy</i>	WCAG 3.1.5
G28	Text does not scale with system font size	All text in the app must remain readable and not overflow or clip when the device system font size is increased up to 200%.	<i>Low Vision / Elderly</i>	WCAG 1.4.4
G29	All-caps text used for body content	All-caps styling must not be used for paragraph text, instructions, or any content longer than a short label.	<i>Cognitive / Dyslexia</i>	WCAG 3.1.5
G30	Icon-only button with no text alternative	Buttons that use only an icon with no visible text label must have a content description that describes the action they perform.	<i>Blind / Low Literacy</i>	WCAG 1.1.1

2. All 30 Detection Rules (R01–R30)

Each rule is a specific, technically detectable check written as: [XML condition] → [flag name].

Rule	Detection Logic (XML Condition → Flag)	Severity	Guidelines
R01	clickable=true AND text="" AND content-desc="" → flag Missing Label	High	G01, G02, G30
R02	class=ImageButton OR clickable ImageView AND content-desc="" → flag No Image Desc	High	G02, G30
R03	Two+ clickable elements share identical text or content-desc with different IDs → flag Duplicate Label	Medium	G03
R04	clickable=true AND (width < 48dp OR height < 48dp) → flag Small Touch Target	High	G04, G17
R05	class=EditText AND hint="" AND text="" AND content-desc="" → flag Unlabeled Input	High	G05, G20
R06	clickable=true AND enabled=false AND no sibling explanation TextView → flag Disabled Control	Medium	G06, G26
R07	bounds width=0 OR height=0 OR [x2<=x1 OR y2<=y1] → flag Zero-Size Element	Medium	G07, G25

R08	Two elements' bounding boxes overlap by >50% of smaller element's area → flag Layout Overlap	Medium	G08, G15
R09	Contrast ratio of foreground vs background < 4.5:1 (normal text) or < 3:1 (large text/icons) → flag Low Contrast	Medium	G09, G11
R10	TextView bounds height < estimated text height at declared textSize → flag Text Overflow	Low/Med	G10, G28, G29
R11	State change shown only via color with no icon or text indicator → flag Color-Only Info	High	G11
R12	VideoView or media player present AND no caption toggle button found → flag Missing Captions	High	G12
R13	AudioView or audio-only media present AND no transcript link/TextView found → flag No Transcript	High	G13
R14	Notification or alert dialog present AND no visible icon or banner element → flag Audio-Only Notification	Medium	G14
R15	TalkBack traversal order does not match visual top-bottom order → flag Bad Focus Order	Medium	G15, G16
R16	Non-interactive decorative element has focusable=true OR importantForAccessibility=yes → flag Decorative In Focus Tree	Low	G16
R17	Distance between two clickable elements < 8dp → flag Insufficient Spacing	Medium	G17
R18	Feature only reachable by gesture with pointer count > 1 AND no single-tap fallback → flag Multi-Gesture Only	High	G18
R19	Single-tap button with text 'delete'/'remove'/'confirm'/'send' AND no confirmation dialog → flag No Confirmation	Medium	G19
R20	EditText has only hint (no paired label TextView) → flag Hint-Only Label	Medium	G05, G20
R21	Error TextView has text="" OR content-desc="" OR does not identify field name → flag Vague Error Message	High	G21
R22	Password EditText (inputType=textPassword) has no adjacent show/hide ImageButton → flag No Password Toggle	Medium	G22
R23	Navigation ImageButton (resource-id contains 'back'/'close'/'home'/'menu') AND content-desc="" → flag Unlabeled Nav Control	High	G01, G23
R24	First or topmost TextView of screen (toolbar title area) has empty text AND empty content-desc → flag Missing Screen Title	Medium	G24
R25	AnimationView or auto-playing media present AND no pause/stop control found → flag Uncontrolled Animation	Medium	G25
R26	Any dialog/countdown TextView present AND no 'Extend'/'OK' button with enabled=true → flag No Timeout Warning	Medium	G26
R27	content-desc or hint contains technical terms (length > 5 words AND avg word length > 8 chars) → flag Complex Label Language	Low	G27
R28	TextView uses sp units AND bounds height does not accommodate 200% font scale → flag Font Scale Overflow	Medium	G10, G28
R29	textAllCaps=true AND text word count > 3 → flag All-Caps Body Text	Low	G29
R30	clickable=true AND text="" AND class contains 'ImageButton'/'ImageView'/'Icon' → flag Icon-Only No Label	High	G01, G30

3. Guideline → Rule Mapping

For each guideline, this table shows which rules implement it.

G-ID	Guideline Issue	Rules Applied	What the Rules Check
G01	Missing accessible label	R01, R23, R30	R01: flag Missing Label R23: flag Unlabeled Nav Control R30: flag Icon-Only No Label
G02	Image button without description	R01, R02	R01: flag Missing Label R02: flag No Image Desc
G03	Duplicate labels	R03	R03: flag Duplicate Label
G04	Small touch target	R04	R04: flag Small Touch Target
G05	Unlabeled input field	R05, R20	R05: flag Unlabeled Input R20: flag Hint-Only Label
G06	Disabled important control	R06	R06: flag Disabled Control
G07	Invisible or zero-size component	R07	R07: flag Zero-Size Element
G08	Possible layout overlap	R08	R08: flag Layout Overlap
G09	Low contrast	R09	R09: flag Low Contrast
G10	Text overflow	R10, R28	R10: flag Text Overflow R28: flag Font Scale Overflow
G11	Information conveyed by color alone	R09, R11	R09: flag Low Contrast R11: flag Color-Only Info
G12	Missing captions on video content	R12	R12: flag Missing Captions
G13	Audio-only content without transcript	R13	R13: flag No Transcript
G14	Notification uses audio only	R14	R14: flag Audio-Only Notification
G15	Logical focus / reading order	R08, R15	R08: flag Layout Overlap R15: flag Bad Focus Order
G16	Non-interactive elements in focus tree	R15, R16	R15: flag Bad Focus Order R16: flag Decorative In Focus Tree
G17	Insufficient touch target spacing	R04, R17	R04: flag Small Touch Target R17: flag Insufficient Spacing
G18	Functionality requires multi-finger gesture	R18	R18: flag Multi-Gesture Only
G19	No confirmation for destructive action	R19	R19: flag No Confirmation
G20	Input label disappears on focus	R05, R20	R05: flag Unlabeled Input R20: flag Hint-Only Label

G21	Vague or missing error messages	R21	R21: flag Vague Error Message
G22	Password field lacks show/hide toggle	R22	R22: flag No Password Toggle
G23	Navigation control not labeled	R23	R23: flag Unlabeled Nav Control
G24	Screen has no descriptive title	R24	R24: flag Missing Screen Title
G25	Uncontrolled auto-updating content	R07, R25	R07: flag Zero-Size Element R25: flag Uncontrolled Animation
G26	Session timeout without warning	R06, R26	R06: flag Disabled Control R26: flag No Timeout Warning
G27	Complex or jargon-heavy labels	R27	R27: flag Complex Label Language
G28	Text does not scale with system font size	R10, R28	R10: flag Text Overflow R28: flag Font Scale Overflow
G29	All-caps text used for body content	R10, R29	R10: flag Text Overflow R29: flag All-Caps Body Text
G30	Icon-only button with no text alternative	R01, R02, R30	R01: flag Missing Label R02: flag No Image Desc R30: flag Icon-Only No Label

4. Rule → Guideline Cross-Reference

For each rule, this table shows which guidelines it enforces.

Rule	Detection Logic	Severity	Enforces
R01	clickable=true AND text="" AND content-desc="" → flag Missing Label	High	G01, G02, G30
R02	class=ImageButton OR clickable ImageView AND content-desc="" → flag No Image Desc	High	G02, G30
R03	Two+ clickable elements share identical text or content-desc with different IDs → flag Duplicate Label	Medium	G03
R04	clickable=true AND (width < 48dp OR height < 48dp) → flag Small Touch Target	High	G04, G17
R05	class=EditText AND hint="" AND text="" AND content-desc="" → flag Unlabeled Input	High	G05, G20
R06	clickable=true AND enabled=false AND no sibling explanation TextView → flag Disabled Control	Medium	G06, G26
R07	bounds width=0 OR height=0 OR [x2<=x1 OR y2<=y1] → flag Zero-Size Element	Medium	G07, G25
R08	Two elements' bounding boxes overlap by >50% of smaller element's area → flag Layout Overlap	Medium	G08, G15

R09	Contrast ratio of foreground vs background < 4.5:1 (normal text) or < 3:1 (large text/icons) → flag Low Contrast	Medium	G09, G11
R10	TextView bounds height < estimated text height at declared textSize → flag Text Overflow	Low/Med	G10, G28, G29
R11	State change shown only via color with no icon or text indicator → flag Color-Only Info	High	G11
R12	VideoView or media player present AND no caption toggle button found → flag Missing Captions	High	G12
R13	AudioView or audio-only media present AND no transcript link/TextView found → flag No Transcript	High	G13
R14	Notification or alert dialog present AND no visible icon or banner element → flag Audio-Only Notification	Medium	G14
R15	TalkBack traversal order does not match visual top-bottom order → flag Bad Focus Order	Medium	G15, G16
R16	Non-interactive decorative element has focusable=true OR importantForAccessibility=yes → flag Decorative In Focus Tree	Low	G16
R17	Distance between two clickable elements < 8dp → flag Insufficient Spacing	Medium	G17
R18	Feature only reachable by gesture with pointer count > 1 AND no single-tap fallback → flag Multi-Gesture Only	High	G18
R19	Single-tap button with text 'delete'/'remove'/'confirm'/'send' AND no confirmation dialog → flag No Confirmation	Medium	G19
R20	EditText has only hint (no paired label TextView) → flag Hint-Only Label	Medium	G05, G20
R21	Error TextView has text="" OR content-desc="" OR does not identify field name → flag Vague Error Message	High	G21
R22	Password EditText (inputType=textPassword) has no adjacent show/hide ImageButton → flag No Password Toggle	Medium	G22
R23	Navigation ImageButton (resource-id contains 'back'/'close'/'home'/'menu') AND content-desc="" → flag Unlabeled Nav Control	High	G01, G23
R24	First or topmost TextView of screen (toolbar title area) has empty text AND empty content-desc → flag Missing Screen Title	Medium	G24
R25	AnimationView or auto-playing media present AND no pause/stop control found → flag Uncontrolled Animation	Medium	G25
R26	Any dialog/countdown TextView present AND no 'Extend'/'OK' button with enabled=true → flag No Timeout Warning	Medium	G26
R27	content-desc or hint contains technical terms (length > 5 words AND avg word length > 8 chars) → flag Complex Label Language	Low	G27
R28	TextView uses sp units AND bounds height does not accommodate 200% font scale → flag Font Scale Overflow	Medium	G10, G28
R29	textAllCaps=true AND text word count > 3 → flag All-Caps Body Text	Low	G29
R30	clickable=true AND text="" AND class contains 'ImageButton'/'ImageView'/'Icon' → flag Icon-Only No Label	High	G01, G30

5. User Group Coverage

This table confirms broad coverage across all disability groups the project targets.

User / Disability Group	Guidelines	Primary Rules
Blind / Screen Reader	G01,G02,G03,G05,G15,G16,G23,G24,G30	R01,R02,R03,R15,R16,R23,R24,R30
Low Vision	G09,G10,G28	R09,R10,R28
Color Blind	G09,G11	R09,R11
Deaf / Hard of Hearing	G12,G13,G14	R12,R13,R14
Motor Impaired / Switch	G04,G06,G17,G18,G19	R04,R06,R17,R18,R19
Elderly	G04,G09,G10,G28	R04,R09,R10,R28
Cognitive / Memory	G03,G20,G21,G22,G27	R03,R20,R21,R22,R27
Cognitive / ADHD / Epilepsy	G25,G26	R25,R26
Dyslexia / Low Literacy	G27,G29	R27,R29
All Users	G06,G07,G08	R06,R07,R08